



Maven Practice

CREATE EC2 INSTANCE



Launch an instance | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Gmail YouTube Contributor Dashbo... OpenAI | ChatGPT Home - Canva Video details - YouT... Programming for Ev... HEC Student Search Student Online - M... HEC - Education Tes... airblue PIA | Great People t... All Bookmarks

aws Services Search [Alt+S]

N. Virginia WaseemKhanDawar

EC2 > Instances > Launch an instance

Success

Successfully initiated launch of instance (i-02fd39c12d38e8424)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

< 1 2 3 4 5 6 >

Create billing and free tier usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Manage detailed monitoring

Create Load Balancer

Create an application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Create AWS budget

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

CloudShell Feedback

© 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



STALL MAVEN IF NOT INSTALLED BY USING THE BELOW COMMAND

➤ YUM INSTALL GIT JAVA-1.8.0-OPENJDK MAVEN -Y

NOTE: ABOVE COMMAND WILL BE INSTALLING 3 PACKAGES **GIT, JAVA AND MAVEN.**



```
44.201.162.237 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...
/home/ec2-user/
Name
..
.ssh
.bash_logout
.bash_profile
.bashrc

Verifying : 1:libglvnd-glx-1.0.1-0.1.git5baa1e5.amzn2.0.1.x86_64
Verifying : guava-13.0-6.amzn2.noarch
Verifying : xalan-j2-2.7.1-23.1.amzn2.noarch
Verifying : objectweb-asm-3.3.1-9.amzn2.noarch
Verifying : junit-4.11-8.1.amzn2.0.1.noarch

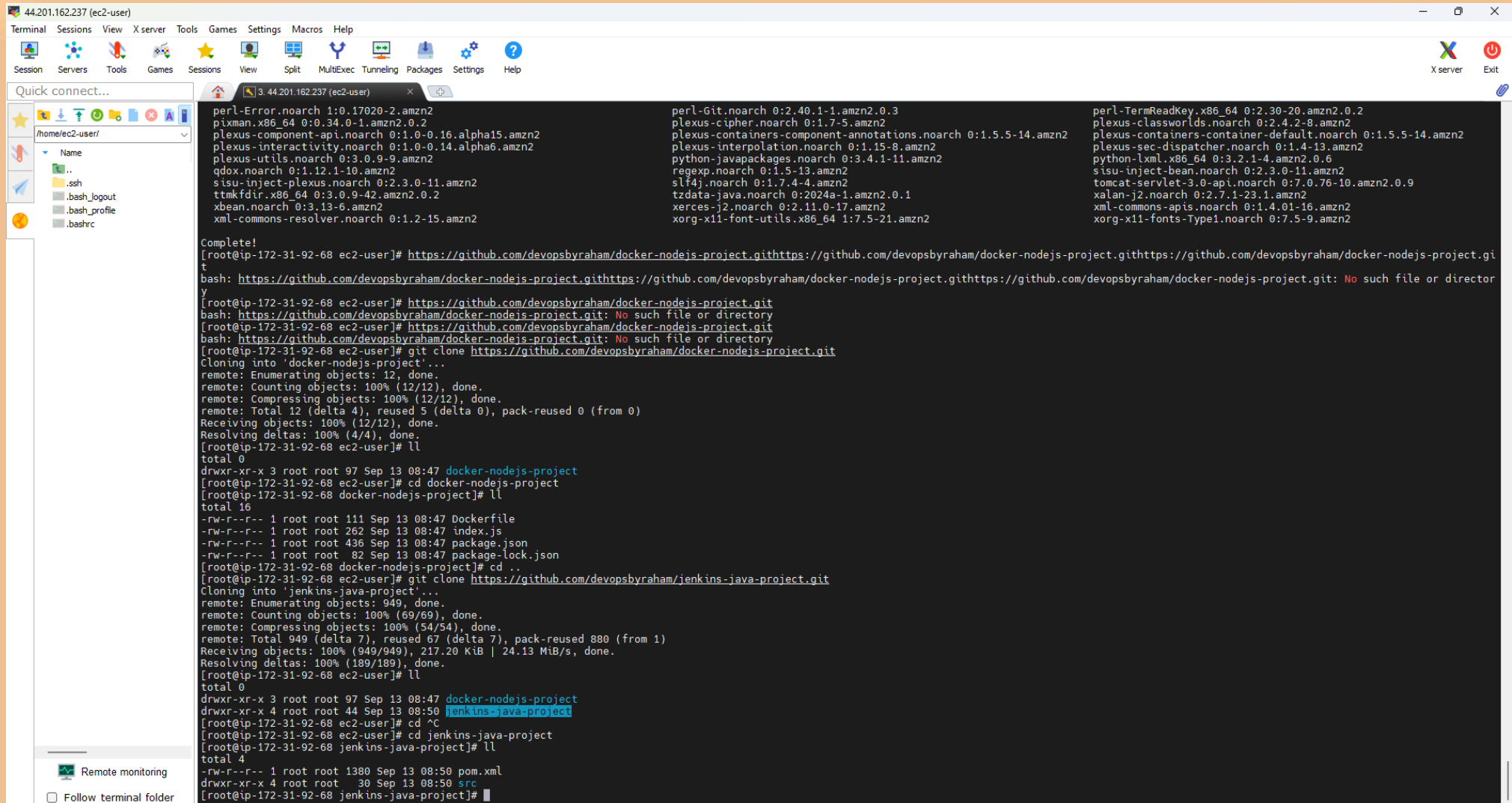
Installed:
git.x86_64 0:2.40.1-1.amzn2.0.3          java-1.8.0-openjdk.x86_64 1:1.8.0.412.b08-1.amzn2.0.1          maven.noarch 0:3.0.5-17.amzn2

Dependency Installed:
aether-api.noarch 0:1.13.1-13.amzn2          aether-connector-wagon.noarch 0:1.13.1-13.amzn2          aether-impl.noarch 0:1.13.1-13.amzn2
aether-spi.noarch 0:1.13.1-13.amzn2          aether-util.noarch 0:1.13.1-13.amzn2          alsa-lib.x86_64 0:1.1.4.1-2.amzn2
aopalliance.noarch 0:1.0-8.1.amzn2          apache-commons-cli.noarch 0:1.2-13.amzn2          apache-commons-codec.noarch 0:1.8-7.amzn2
apache-commons-io.noarch 1:2.4-12.amzn2.0.1  apache-commons-lang.noarch 0:2.6-15.amzn2          apache-commons-logging.noarch 0:1.1.2-7.amzn2
apache-commons-net.noarch 0:3.2-8.amzn2       atinject.noarch 0:1-13.20100611svn86.amzn2          apache-commons-pool.noarch 0:3.10-1.amzn2
avahi-libs.x86_64 0:0.6.31-20.amzn2.0.5      avalon-framework.noarch 0:4.3-10.1.amzn2          atk.x86_64 0:2.22.0-3.amzn2.0.2
bcel.noarch 0:5.2-18.amzn2.0.1               cairo.x86_64 0:1.15.12-4.amzn2          avalon-logkit.noarch 0:2.1-14.1.amzn2
cdi-api.noarch 0:1.0-11.SP4.amzn2             cglib.noarch 0:2.2-18.1.amzn2          cal10n.noarch 0:0.7.4.amzn2
cups-libs.x86_64 1:1.6.3-51.amzn2.0.5        dejavu-fonts-common.noarch 0:2.33-6.amzn2          copy-jdk-configs.noarch 0:3.3-10.amzn2
easymock2.noarch 0:2.5.2-12.1.amzn2           felix-framework.noarch 0:4.2.1-5.amzn2          dejavu-sans-fonts.noarch 0:2.33-6.amzn2
fontpackages-filesystem.noarch 0:1.44-8.amzn2  friditl.x86_64 0:1.0.2-1.amzn2.1.2          fontconfig.x86_64 0:2.13.0-4.3.amzn2
geronimo-annotation.noarch 0:1.0-15.amzn2     geronimo-jms.noarch 0:1.1.1-19.amzn2          gdk-pixbuf2.x86_64 0:2.36.12-3.amzn2
git-core.x86_64 0:2.40.1-1.amzn2.0.3          git-core-doc.noarch 0:2.40.1-1.amzn2.0.3          giflib.x86_64 0:4.1.6-9.amzn2.0.2
graphite2.x86_64 0:1.3.10-1.amzn2.0.2         gtk-update-icon-cache.x86_64 0:3.22.30-3.amzn2.0.1  google-guice.noarch 0:3.1.3-9.amzn2
guava.noarch 0:13.0-6.amzn2                  hamcrest.noarch 0:1.3-6.1.amzn2          gtk2.x86_64 0:2.24.31-1.amzn2.0.3
hicolor-icon-theme.noarch 0:0.12-7.amzn2       httpcomponents-client.noarch 0:4.2.5-5.amzn2.0.1  harfbuzz.x86_64 0:1.7.5-2.amzn2.0.2
jakarta-commons-httpclient.noarch 1:3.1-16.amzn2  httpcomponents-core.noarch 0:4.2.4-6.amzn2          httpcomponents-core.noarch 0:4.2.4-6.amzn2
java-1.8.0-openjdk-headless.x86_64 1:1.8.0.412.b08-1.amzn2.0.1  jasper-libs.x86_64 0:1.900.1-33.amzn2.0.1          java-1.8.0-openjdk-devel.x86_64 1:1.8.0.412.b08-1.amzn2.0.1
javassist.noarch 0:3.16.1-10.amzn2           javamail.noarch 0:1.4.6-8.amzn2          javapackages-tools.noarch 0:3.4.1-11.amzn2
jboss-interceptors-1.1-api.noarch 0:1.0.2-0.9.20120319git49a904.amzn2  jboss-ejb-3.1-api.noarch 0:1.0.2-10.amzn2          jboss-el-2.2-api.noarch 0:1.0.1-0.7.20120212git2fabd8.amzn2
jboss-interceptors-1.1-api.noarch 0:1.0.2-0.9.20120319git49a904.amzn2  jboss-jaxrpc-1.1-api.noarch 0:1.0.1-7.amzn2          jboss-servlet-3.0-api.noarch 0:1.0.1-9.amzn2
jboss-transaction-1.1-api.noarch 0:1.0.1-8.amzn2  jline.noarch 0:1.0-8.amzn2          jsch.noarch 0:0.1.50-5.amzn2
jsoup.noarch 0:1.6.1-10.amzn2.0.1            junit.noarch 0:4.11-8.1.amzn2.0.1          jzlib.noarch 0:1.1.1-6.amzn2
libICE.x86_64 0:1.0.9-9.amzn2.0.2            libSM.x86_64 0:1.2.2-2.amzn2.0.2          libX11.x86_64 0:1.6.7-3.amzn2.0.5
libX11-common.noarch 0:1.6.7-3.amzn2.0.5      libXau.x86_64 0:1.0.8-2.1.amzn2.0.2          libXcomposite.x86_64 0:0.4.4-4.1.amzn2.0.2
libXcursor.x86_64 0:1.1.15-1.amzn2           libXdamage.x86_64 0:1.1.4-4.1.amzn2.0.2          libXext.x86_64 0:1.3.3-3.amzn2.0.2
libXfixes.x86_64 0:5.0.3-1.amzn2.0.2         libXft.x86_64 0:2.3.2-2.amzn2.0.2          libXi.x86_64 0:1.7.9-1.amzn2.0.2
libXinerama.x86_64 0:1.1.3-2.1.amzn2.0.2     libXrandr.x86_64 0:1.5.1-2.amzn2.0.3          libXrender.x86_64 0:0.9.10-1.amzn2.0.2
libXtst.x86_64 0:1.2.3-1.amzn2.0.2           libXxf86vm.x86_64 0:1.1.4-1.amzn2.0.2          libfontenc.x86_64 0:1.1.3-3.amzn2.0.2
libglvnd.x86_64 1:1.0.1-0.1.git5baa1e5.amzn2.0.1  libglvnd-egl.x86_64 1:1.0.1-0.1.git5baa1e5.amzn2.0.1  libglvnd-glx.x86_64 1:1.0.1-0.1.git5baa1e5.amzn2.0.1
libthai.x86_64 0:0.1.14-9.amzn2.0.2          libwayland-client.x86_64 0:1.17.0-1.amzn2.0.1          libwayland-server.x86_64 0:1.17.0-1.amzn2.0.1
libxcb.x86_64 0:1.12-1.amzn2.0.2            libxshmfence.x86_64 0:1.2-1.amzn2.0.2          libxslt.x86_64 0:1.1.28-6.amzn2
lkscpt-tools.x86_64 0:1.0.17-2.amzn2.0.2      log4j.noarch 0:1.2.17-18.amzn2          log4j-cve-2021-44228-hotpatch.noarch 0:1.3-7.amzn2
maven-wagon.noarch 0:2.4-3.amzn2             mesa-libEGL.x86_64 0:18.3.4-5.amzn2.0.1          mesa-libGL.x86_64 0:18.3.4-5.amzn2.0.1
mesa-libgbm.x86_64 0:18.3.4-5.amzn2.0.1      mesa-libglapi.x86_64 0:18.3.4-5.amzn2.0.1          nekohtml.noarch 0:1.9.14-13.amzn2
objectweb-asm.noarch 0:3.3.1-9.amzn2          perl-Error.noarch 1:0.17020-2.amzn2          pascalite-libs.x86_64 0:1.8.8-7.amzn2
perl-Error.noarch 1:0.17020-2.amzn2           perl-Git.noarch 0:2.40.1-1.amzn2.0.3          perl-TermReadKey.x86_64 0:2.30-20.amzn2.0.2
pixmap.x86_64 0:0.34.0-1.amzn2.0.2           plexus-cipher.noarch 0:1.7-5.amzn2          plexus-classworlds.noarch 0:2.4.2-8.amzn2
plexus-component-api.noarch 0:1.0.0-16.alpha15.amzn2  plexus-containers-component-annotations.noarch 0:1.5.5-14.amzn2  plexus-containers-container-default.noarch 0:1.5.5-14.amzn2
plexus-interactivity.noarch 0:1.0.0-14.alpha6.amzn2  plexus-interpolation.noarch 0:1.15-8.amzn2          plexus-sec-dispatcher.noarch 0:1.4.13.amzn2
plexus-utils.noarch 0:3.0.9-9.amzn2           python-javapackages.noarch 0:3.4.1-11.amzn2          python-lxml.x86_64 0:3.2.1-4.amzn2.0.6
qdox.noarch 0:1.12.1-10.amzn2               regexp.noarch 0:1.5-13.amzn2          python-mako.noarch 0:2.3.0-11.amzn2
sisu-inject-plexus.noarch 0:2.3.0-11.amzn2    slf4j.noarch 0:1.7.4-4.amzn2          tomcat-servlet-3.0-api.noarch 0:7.0.76-10.amzn2.0.9
ttmkfdir.x86_64 0:3.0.9-42.amzn2.0.2         tzdata-java.noarch 0:2024a-1.amzn2.0.1          xalan-j2.noarch 0:2.7.1-23.1.amzn2
xbean.noarch 0:3.13-6.amzn2                  xerces-j2.noarch 0:2.11.0-17.amzn2          xml-commons-apis.noarch 0:1.4.01-16.amzn2
xml-commons-resolver.noarch 0:1.2-15.amzn2     xorg-x11-font-utils.x86_64 1:7.5-21.amzn2          xorg-x11-fonts-Type1.noarch 0:7.5-9.amzn2

Complete!
[root@ip-172-31-92-68 ec2-user]#
```

NOW BY USING GIT CLONE COMMAND TO CLONE THE JAVA PROGRAM FROM THE GITHUB LINK:

➤ git clone <https://github.com/devopsbyraham/jenkins-java-project.git>



```
44.201.162.237 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/home/ec2-user/
Name
..
.ssh
.bash_logout
.bash_profile
.bashrc
perl-Error.noarch 1:0.17020-2.amzn2
pixmap.x86_64 0:0.34.0-1.amzn2.0.2
plexus-component-api.noarch 0:1.0-0.16.alpha15.amzn2
plexus-interactive.noarch 0:1.0-0.14.alpha6.amzn2
plexus-utils.noarch 0:3.0.9-9.amzn2
qdox.noarch 0:1.12.1-10.amzn2
sisu-inject-plexus.noarch 0:2.3.0-11.amzn2
ttmkfdir.x86_64 0:3.0.9-42.amzn2.0.2
xbean.noarch 0:3.13-6.amzn2
xml-commons-resolver.noarch 0:1.2-15.amzn2
perl-Git.noarch 0:2.40.1-1.amzn2.0.3
plexus-cipher.noarch 0:1.7-5.amzn2
plexus-containers-component-annotations.noarch 0:1.5.5-14.amzn2
plexus-interpolation.noarch 0:1.15-8.amzn2
python-javapackages.noarch 0:3.4.1-11.amzn2
regex.noarch 0:1.5-13.amzn2
slf4j.noarch 0:1.7.4-4.amzn2
tzdata-java.noarch 0:2024a-1.amzn2.0.1
xerces-j2.noarch 0:2.11.0-17.amzn2
xorg-x11-font-utils.x86_64 1:7.5-21.amzn2
perl-TermReadKey.x86_64 0:2.30-20.amzn2.0.2
plexus-classworlds.noarch 0:2.4.2-8.amzn2
plexus-containers-container-default.noarch 0:1.5.5-14.amzn2
plexus-sec-dispatcher.noarch 0:1.4-13.amzn2
python-lxml.x86_64 0:3.2.1-4.amzn2.0.6
sisu-inject-bean.noarch 0:2.3.0-11.amzn2
tomcat-servlet-3.0-api.noarch 0:7.0.76-10.amzn2.0.9
xalan-j2.noarch 0:2.7.1-23.1.amzn2
xml-commons-apis.noarch 0:1.4.01-16.amzn2
xorg-x11-fonts-Type1.noarch 0:7.5-9.amzn2
Complete!
[root@ip-172-31-92-68 ec2-user]# https://github.com/devopsbyraham/docker-nodejs-project.githttps://github.com/devopsbyraham/docker-nodejs-project.githttps://github.com/devopsbyraham/docker-nodejs-project.git
bash: https://github.com/devopsbyraham/docker-nodejs-project.githttps://github.com/devopsbyraham/docker-nodejs-project.githttps://github.com/devopsbyraham/docker-nodejs-project.git: No such file or director
y
[root@ip-172-31-92-68 ec2-user]# https://github.com/devopsbyraham/docker-nodejs-project.git
bash: https://github.com/devopsbyraham/docker-nodejs-project.git: No such file or directory
[root@ip-172-31-92-68 ec2-user]# https://github.com/devopsbyraham/docker-nodejs-project.git
bash: https://github.com/devopsbyraham/docker-nodejs-project.git: No such file or directory
[root@ip-172-31-92-68 ec2-user]# git clone https://github.com/devopsbyraham/docker-nodejs-project.git
Cloning into 'docker-nodejs-project'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (12/12), done.
remote: Total 12 (delta 4), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), done.
Resolving deltas: 100% (4/4), done.
[root@ip-172-31-92-68 ec2-user]# ll
total 0
drwxr-xr-x 3 root root 97 Sep 13 08:47 docker-nodejs-project
[root@ip-172-31-92-68 ec2-user]# cd docker-nodejs-project
[root@ip-172-31-92-68 docker-nodejs-project]# ll
total 16
-rw-r--r-- 1 root root 111 Sep 13 08:47 Dockerfile
-rw-r--r-- 1 root root 262 Sep 13 08:47 index.js
-rw-r--r-- 1 root root 436 Sep 13 08:47 package.json
-rw-r--r-- 1 root root 82 Sep 13 08:47 package-lock.json
[root@ip-172-31-92-68 docker-nodejs-project]# cd ..
[root@ip-172-31-92-68 ec2-user]# git clone https://github.com/devopsbyraham/jenkins-java-project.git
Cloning into 'jenkins-java-project'...
remote: Enumerating objects: 949, done.
remote: Counting objects: 100% (69/69), done.
remote: Compressing objects: 100% (54/54), done.
remote: Total 949 (delta 7), reused 67 (delta 7), pack-reused 880 (from 1)
Receiving objects: 100% (949/949), 217.20 KiB | 24.13 MiB/s, done.
Resolving deltas: 100% (189/189), done.
[root@ip-172-31-92-68 ec2-user]# ll
total 0
drwxr-xr-x 3 root root 97 Sep 13 08:47 docker-nodejs-project
drwxr-xr-x 4 root root 44 Sep 13 08:50 jenkins-java-project
[root@ip-172-31-92-68 ec2-user]# cd ^C
[root@ip-172-31-92-68 ec2-user]# cd jenkins-java-project
[root@ip-172-31-92-68 jenkins-java-project]# ll
total 4
-rw-r--r-- 1 root root 1380 Sep 13 08:50 pom.xml
drwxr-xr-x 4 root root 30 Sep 13 08:50 src
[root@ip-172-31-92-68 jenkins-java-project]#
```

OPTIONAL: INSTALL NEW PACKAGE CALLED TREE BY USING THE COMMAND

➤ `yum install tree`

NOTE: IT WILL HELP US TO UNDERSTAND THE WORK OF MAVEN COMMAND AND WILL MAKE IT EASY TO UNDERSTAND.



```
52.70.53.188 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/home/ec2-user/
Name
..
.ssh
docker-nodejs-project
jenkins-java-project
.bash_history
.bash_logout
.bash_profile
.bashrc
Is this ok [y/d/N]: y
Downloading packages:
tree-1.6.0-10.amzn2.0.1.x86_64.rpm | 47 kB 00:00:00
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Installing : tree-1.6.0-10.amzn2.0.1.x86_64 1/1
Verifying : tree-1.6.0-10.amzn2.0.1.x86_64 1/1

Installed:
tree.x86_64 0:1.6.0-10.amzn2.0.1

Complete!
[root@ip-172-31-92-68 jenkins-java-project]# tree
.
├── pom.xml
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── in
│   │   │   │   ├── javahome
│   │   │   │   │   ├── myweb
│   │   │   │   │   │   ├── controller
│   │   │   │   │   │   │   Calculator.java
│   │   │   ├── webapp
│   │   │   │   ├── index.jsp
│   │   │   │   ├── style.css
│   │   │   │   ├── WEB-INF
│   │   │   │   │   └── web.xml
│   │   └── test
│   │       ├── java
│   │       │   ├── in
│   │       │   │   ├── javahome
│   │       │   │   │   ├── myweb
│   │       │   │   │   │   ├── controller
│   │       │   │   │   │   │   CalculatorTest.java
└──
```

15 directories, 6 files
[root@ip-172-31-92-68 jenkins-java-project]#



NOTES:

THE INSTALLATION OF THE TREE PACKAGE IS GENERALLY MEANT FOR VIEWING THE DIRECTORY AND FILE STRUCTURES GRAPHICALLY. IN RELATION TO MAVEN, YOU CAN USE THE TREE COMMAND TO SEE HOW MANY OTHER MAVEN COMMANDS, FROM MVN COMPILE TO MVN CLEAN, CHANGE THE FILE STRUCTURE AND CONTENTS OF OUR PROJECT. HERE'S HOW IT'S DONE:

- BEFORE EXECUTING MAVEN COMMANDS: RUN TREE BEFORE RUNNING ANY MAVEN COMMAND TO TAKE THE INITIAL PROJECT STRUCTURE.**
- AFTER EXECUTING MAVEN COMMANDS: AFTER PERFORMING EACH MAVEN COMMAND, TREE AGAIN CAN BE USED TO SEE WHICH FILES OR DIRECTORIES WERE CREATED, DELETED, OR MODIFIED. EVERYTHING WILL BE CLEAR AND VISUALLY PERCEIVED IN A VIEW HOW MAVEN MANAGES THE BUILD PROCESS AND OUTPUT FILES.**

FOR EXAMPLE:

- MVN COMPILE: PACKS COMPILED .CLASS FILES IN THE TARGET DIRECTORY.**
- MVN TEST: RUNS TESTS AND PERHAPS GENERATES REPORTS OR TEST-RELATED FILES.**
- MVN PACKAGE: PUTS THE COMPILED CODE INTO A JAR, BOTH WITHIN THE TARGET DIRECTORY.**
- MVN INSTALL: INSTALLS THE PACKAGED .JAR FILE INTO THE LOCAL MAVEN REPOSITORY.**
- MVN CLEAN: CLEANS UP THE TARGET DIRECTORY, WHERE ALL BUILD ARTIFACTS ARE ERASED.**
- MVN CLEAN PACKAGE: CLEANSSES TARGET DIRECTORY AFTER REPACKAGING THE PROJECT.**

USING A TREE COMMAND BEFORE AND AFTER THESE MAVEN COMMANDS LETS US SEE EXACTLY WHERE THE FILES WERE CREATED, EXACTLY WHAT WAS CREATED, AND AT JUST WHICH POINTS IN THE LIFECYCLE OF THE PROJECT STRUCTURE CHANGES.

1ST COMMAND OF MAVEN (MVN COMPILE):

THIS COMMAND USES TO COMPILE THE CODE AND COVERT THE CODE FROM .JAVA TO .CLASS

THE COMMAND IS:

➤ mvn compile



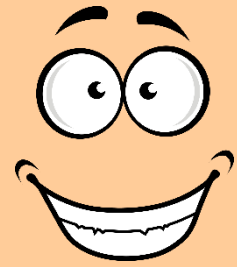
```
52.70.53.188 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/home/ec2-user/
Name
ssh
docker-nodejs-project
jenkins-java-project
.bash_history
.bash_logout
.bash_profile
.bashrc
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/1.8.1/plexus-compiler-api-1.8.1.pom (805 B at 78.6 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-compiler/1.8.1/plexus-compiler-1.8.1.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler/1.8.1/plexus-compiler-1.8.1.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler/1.8.1/plexus-compiler-1.8.1.pom (4 KB at 377.3 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-components/1.1.18/plexus-components-1.1.18.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-components/1.1.18/plexus-components-1.1.18.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-components/1.1.18/plexus-components-1.1.18.pom (6 KB at 475.4 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus/2.0.7/plexus-2.0.7.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/2.0.7/plexus-2.0.7.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/2.0.7/plexus-2.0.7.pom (17 KB at 1407.5 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-compiler-manager/1.8.1/plexus-compiler-manager-1.8.1.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/1.8.1/plexus-compiler-manager-1.8.1.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/1.8.1/plexus-compiler-manager-1.8.1.pom (713 B at 58.0 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-compiler-javac/1.8.1/plexus-compiler-javac-1.8.1.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-javac/1.8.1/plexus-compiler-javac-1.8.1.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-javac/1.8.1/plexus-compiler-javac-1.8.1.pom (710 B at 53.3 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-compilers/1.8.1/plexus-compilers-1.8.1.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compilers/1.8.1/plexus-compilers-1.8.1.pom (2 KB at 161.9 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-compiler-api/1.8.1/plexus-compiler-api-1.8.1.jar
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-compiler-manager/1.8.1/plexus-compiler-manager-1.8.1.jar
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-compiler-javac/1.8.1/plexus-compiler-javac-1.8.1.jar
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/1.8.1/plexus-compiler-api-1.8.1.jar
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/1.8.1/plexus-compiler-manager-1.8.1.jar (6 KB at 472.7 KB/sec)
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-javac/1.8.1/plexus-compiler-javac-1.8.1.jar (13 KB at 1111.6 KB/sec)
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/ec2-user/jenkins-java-project/target/classes
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 6.819s
[INFO] Finished at: Tue Sep 17 05:25:15 UTC 2024
[INFO] Final Memory: 15M/37M
[INFO] -----
[root@ip-172-31-92-68 jenkins-java-project]#
```


2ND COMMAND OF MAVEN (MVN TEST):

THIS COMMAND USED TO CHECK THE ERRORS IN THE GIVEN JAVA PROGRAM.

THE COMMAND IS:

➤ mvn test



```
52.70.53.188 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/home/ec2-user/
Name
.ssh
docker-nodejs-project
jenkins-java-project
.bash_history
.bash_logout
.bash_profile
.bashrc

Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/2.1/plexus-utils-2.1.jar
Downloaded: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-booter/2.10/surefire-booter-2.10.jar (34 KB at 1322.4 KB/sec)
Downloading: https://repo.maven.apache.org/maven2/org/apache/maven/reporting/maven-reporting-api/2.0.9/maven-reporting-api-2.0.9.jar
Downloaded: https://repo.maven.apache.org/maven2/org/apache/maven/reporting/maven-reporting-api/2.0.9/maven-reporting-api-2.0.9.jar (10 KB at 495.0 KB/sec)
Downloaded: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-common-artifact-filters/1.3/maven-common-artifact-filters-1.3.jar (31 KB at 473.9 KB/sec)
Downloaded: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/maven-surefire-common/2.10/maven-surefire-common-2.10.jar (60 KB at 796.3 KB/sec)
Downloaded: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-api/2.10/surefire-api-2.10.jar (158 KB at 1694.0 KB/sec)
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/2.1/plexus-utils-2.1.jar (220 KB at 2237.1 KB/sec)
[INFO] Surefire report directory: /home/ec2-user/jenkins-java-project/target/surefire-reports
Downloading: http://repo1.maven.org/maven2/org/apache/maven/surefire/surefire-junit3/2.10/surefire-junit3-2.10.pom
Downloaded: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit3/2.10/surefire-junit3-2.10.pom (2 KB at 185.0 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/apache/maven/surefire/surefire-providers/2.10/surefire-providers-2.10.pom
Downloaded: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-providers/2.10/surefire-providers-2.10.pom (3 KB at 201.2 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/apache/maven/surefire/surefire-junit3/2.10/surefire-junit3-2.10.jar
Downloaded: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit3/2.10/surefire-junit3-2.10.jar (26 KB at 1701.9 KB/sec)

-----
T E S T S
-----
Running in.javahome.myweb.controller.CalculatorTest
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.02 sec

Results :

Tests run: 2, Failures: 0, Errors: 0, Skipped: 0

[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.787s
[INFO] Finished at: Tue Sep 17 05:29:11 UTC 2024
[INFO] Final Memory: 16M/39M
[INFO] -----
[root@ip-172-31-92-68 jenkins-java-project]#
```

THIS COMMAND WILL USE TO CONVERT .CLASS FILE TO .JAR FILE WITHIN THE TARGET DIRECTORY

THE COMMAND IS:

➤ mvn package



```
52.70.53.188 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...
/home/ec2-user/
Name
..
.ssh
docker-nodejs-project
jenkins-java-project
.bash_history
.bash_logout
.bash_profile
.bashrc

Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/1.5.6/plexus-utils-1.5.6.pom (6 KB at 517.8 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus/1.0.12/plexus-1.0.12.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/1.0.12/plexus-1.0.12.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/1.0.12/plexus-1.0.12.pom (10 KB at 683.6 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-interpolation/1.6/plexus-interpolation-1.6.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.6/plexus-interpolation-1.6.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.6/plexus-interpolation-1.6.pom (3 KB at 258.1 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/apache/maven/maven-archiver/2.4.1/maven-archiver-2.4.1.jar
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-archiver/1.2/plexus-archiver-1.2.jar
Downloading: http://repo1.maven.org/maven2/com/thoughtworks/xstream/xstream/1.3.1/xstream-1.3.1.jar
Downloading: http://repo1.maven.org/maven2/org/apache/maven/shared/maven-filtering/1.0-beta-2/maven-filtering-1.0-beta-2.jar
Downloading: http://repo1.maven.org/maven2/xpp3/xpp3_min/1.1.4c/xpp3_min-1.1.4c.jar
Downloading: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/2.4.1/maven-archiver-2.4.1.jar
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/1.2/plexus-archiver-1.2.jar
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/1.0.1/plexus-io-1.0.1.jar
Downloaded: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/2.4.1/maven-archiver-2.4.1.jar (20 KB at 934.6 KB/sec)
Downloading: https://repo.maven.apache.org/maven2/com/thoughtworks/xstream/xstream/1.3.1/xstream-1.3.1.jar
Downloading: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-filtering/1.0-beta-2/maven-filtering-1.0-beta-2.jar
Downloading: https://repo.maven.apache.org/maven2/xpp3/xpp3_min/1.1.4c/xpp3_min-1.1.4c.jar
Downloaded: https://repo.maven.apache.org/maven2/xpp3/xpp3_min/1.1.4c/xpp3_min-1.1.4c.jar (25 KB at 427.6 KB/sec)
Downloaded: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-filtering/1.0-beta-2/maven-filtering-1.0-beta-2.jar (33 KB at 549.2 KB/sec)
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/1.0.1/plexus-io-1.0.1.jar (50 KB at 598.5 KB/sec)
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/1.2/plexus-archiver-1.2.jar (178 KB at 1585.5 KB/sec)
Downloaded: https://repo.maven.apache.org/maven2/com/thoughtworks/xstream/xstream/1.3.1/xstream-1.3.1.jar (422 KB at 3695.6 KB/sec)
[INFO] Packaging webapp
[INFO] Assembling webapp [NETFLIX] in [/home/ec2-user/jenkins-java-project/target/NETFLIX-1.2.2]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/ec2-user/jenkins-java-project/src/main/webapp]
[INFO] Webapp assembled in [99 msecs]
[INFO] Building war: /home/ec2-user/jenkins-java-project/target/NETFLIX-1.2.2.war
[INFO] WEB-INF/web.xml already added, skipping
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.902s
[INFO] Finished at: Tue Sep 17 05:34:33 UTC 2024
[INFO] Final Memory: 11M/28M
[INFO] -----
[root@ip-172-31-92-68 jenkins-java-project]#
```



4TH COMMAND OF MAVEN (MVN INSTALL):

THIS COMMAND WILL USE TO CONVERT .CLASS FILE TO .JAR FILE TO THE LOCAL MAVEN DIRECTORY

THE COMMAND IS:

➤ mvn install

```
52.70.53.188 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/home/ec2-user/
Name
..
.ssh
docker-nodejs-project
jenkins-java-project
.bash_history
.bash_logout
.bash_profile
.bashrc

Results :
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0

[INFO] --- maven-war-plugin:2.1.1:war (default-war) @ NETFLIX ---
[INFO] Packaging webapp
[INFO] Assembling webapp [NETFLIX] in [/home/ec2-user/jenkins-java-project/target/NETFLIX-1.2.2]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/ec2-user/jenkins-java-project/src/main/webapp]
[INFO] Webapp assembled in [155 msecs]
[INFO] Building war: /home/ec2-user/jenkins-java-project/target/NETFLIX-1.2.2.war
[INFO] WEB-INF/web.xml already added, skipping
[INFO] --- maven-install-plugin:2.3.1:install (default-install) @ NETFLIX ---
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.pom (2 KB at 86.2 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-components/1.1.7/plexus-components-1.1.7.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-components/1.1.7/plexus-components-1.1.7.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-components/1.1.7/plexus-components-1.1.7.pom (5 KB at 323.9 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus/1.0.8/plexus-1.0.8.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/1.0.8/plexus-1.0.8.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/1.0.8/plexus-1.0.8.pom (8 KB at 294.2 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-container-default/1.0-alpha-8/plexus-container-default-1.0-alpha-8.pom
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-container-default/1.0-alpha-8/plexus-container-default-1.0-alpha-8.pom
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-container-default/1.0-alpha-8/plexus-container-default-1.0-alpha-8.pom (8 KB at 506.6 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar (12 KB at 772.0 KB/sec)
[INFO] Installing /home/ec2-user/jenkins-java-project/target/NETFLIX-1.2.2.war to /root/.m2/repository/in/RAHAM/NETFLIX/1.2.2/NETFLIX-1.2.2.war
[INFO] Installing /home/ec2-user/jenkins-java-project/pom.xml to /root/.m2/repository/in/RAHAM/NETFLIX/1.2.2/NETFLIX-1.2.2.pom
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.473s
[INFO] Finished at: Tue Sep 17 05:38:59 UTC 2024
[INFO] Final Memory: 11M/28M
```



5TH COMMAND OF MAVEN (MVN CLEAN PACKAGE): THE MOST IMPORTANT MAVEN COMMAND THAT PERFORMS ALL THE STEPS (FROM COMPILE TO INSTALL) IN ONE GO IS:

➤ mvn install

```
52.70.53.188 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/home/ec2-user/
Name
..
.ssh
docker-nodejs-project
jenkins-java-project
.bash_history
.bash_logout
.bash_profile
.bashrc
t 506.6 KB/sec)
Downloading: http://repo1.maven.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar
Downloading: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar
Downloaded: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar (12 KB at 772.0 KB/sec)
[INFO] Installing /home/ec2-user/jenkins-java-project/target/NETFLIX-1.2.2.war to /root/.m2/repository/in/RAHAM/NETFLIX/1.2.2/NETFLIX-1.2.2.war
[INFO] Installing /home/ec2-user/jenkins-java-project/pom.xml to /root/.m2/repository/in/RAHAM/NETFLIX/1.2.2/NETFLIX-1.2.2.pom
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.473s
[INFO] Finished at: Tue Sep 17 05:38:59 UTC 2024
[INFO] Final Memory: 11M/28M
[INFO] -----
[root@ip-172-31-92-68 jenkins-java-project]# cd /root/.m2/repository/in/RAHAM/NETFLIX/1.2.2/NETFLIX-1.2.2.pom
bash: cd: /root/.m2/repository/in/RAHAM/NETFLIX/1.2.2/NETFLIX-1.2.2.pom: Not a directory
[root@ip-172-31-92-68 jenkins-java-project]# ll
total 4
-rw-r--r-- 1 root root 1380 Sep 13 08:50 pom.xml
drwxr-xr-x 4 root root 30 Sep 13 08:50 src
drwxr-xr-x 8 root root 149 Sep 17 05:34 target
[root@ip-172-31-92-68 jenkins-java-project]# cd /root/.m2/repository/in/RAHAM/NETFLIX/1.2.2/NETFLIX-1.2.2.pom/
bash: cd: /root/.m2/repository/in/RAHAM/NETFLIX/1.2.2/NETFLIX-1.2.2.pom/: Not a directory
[root@ip-172-31-92-68 jenkins-java-project]# cd 1.2.2
bash: cd: 1.2.2: No such file or directory
[root@ip-172-31-92-68 jenkins-java-project]# ll
total 4
-rw-r--r-- 1 root root 1380 Sep 13 08:50 pom.xml
drwxr-xr-x 4 root root 30 Sep 13 08:50 src
drwxr-xr-x 8 root root 149 Sep 17 05:34 target
[root@ip-172-31-92-68 jenkins-java-project]# cd..
bash: cd.: command not found
[root@ip-172-31-92-68 jenkins-java-project]# ls
pom.xml src target
[root@ip-172-31-92-68 jenkins-java-project]# cd /root/.m2/repository/in/RAHAM/NETFLIX/1.2.2/
[root@ip-172-31-92-68 1.2.2]# ll
total 8380
-rw-r--r-- 1 root root 166 Sep 17 05:38 _maven.repositories
-rw-r--r-- 1 root root 1380 Sep 13 08:50 NETFLIX-1.2.2.pom
-rw-r--r-- 1 root root 8569330 Sep 17 05:38 NETFLIX-1.2.2.war
[root@ip-172-31-92-68 1.2.2]#
```


THE ABOVE COMMAND PERFORMS THE FOLLOWING TASKS IN SEQUENCE:

- 1. COMPILES THE CODE (MVN COMPILE)**
- 2. RUNS THE TESTS (MVN TEST)**
- 3. PACKAGES THE CODE INTO A .JAR OR .WAR FILE (MVN PACKAGE)**
- 4. INSTALLS THE PACKAGED CODE INTO THE LOCAL MAVEN REPOSITORY (MVN INSTALL)**

IN SUMMARY, MVN INSTALL EXECUTES ALL THE KEY STEPS IN ONE COMMAND, MAKING IT UNNECESSARY TO RUN MVN COMPILE, MVN TEST, MVN PACKAGE, OR MVN INSTALL SEPARATELY.

6TH COMMAND OF MAVEN (MVN CLEAN):

DELETE THE TARGET DIRECTORY, WHERE ALL BUILD ARTIFACTS ARE ERASED.

THE COMMAND IS

➤ **mvn clean**

PRACTICE AND CHECK THE OUTPUT OF THIS COMMAND!!

CREATED BY,

WASEEM KHAN DAWAR

